

knowledge of critical wetland ecosystem processes is fragmented and incomplete. The MERP study was designed to overcome the difficulties associated with the study of natural wetlands. Although the impoundments used were artificial, the MERP experiments were well designed. The results confirmed several aspects of wetland ecology observed by early investigators and provided new information on factors that influence the productivity of prairie wetlands.

The MERP experiments used an interdisciplinary approach to simultaneously evaluate major components (nutrient budgets, macrophytes, algae, invertebrates, and vertebrates) of the prairie wetland ecosystem. The studies demonstrated the importance of macrophytes and algae to wetland productivity and did much to clarify the complex ecological relationships involving other taxonomic groups and processes. Another important result was that the diversity of habitat features changing over time from climate variations was more important than a particular wetland phase in maintaining diverse biological communities.

The main drawback with the MERP experiments was that only semipermanently flooded wetlands were evaluated; other wetland classes in the prairies were not evaluated nor were the various hydrologic processes that make specific wetland classes, and their biological communities, unique. Nevertheless, the authors and editors did an excellent job organizing the valuable information generated from the MERP experiments and synthesizing it within what is known of prairie wetlands, including other wetland classes. This book will be an excellent source for our state of knowledge of prairie wetlands for some time to come. It should be required reading for students of wetland ecology, and I believe it will become a standard reference for veteran researchers.

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## NEURAL SCIENCES

HOW BRAINS MAKE UP THEIR MINDS. *Maps of the Mind.*

By Walter J Freeman. New York: Columbia University Press. \$24.95. ix + 171 p; ill.; index. ISBN: 0-231-12008-7. 2000.

The author may be one of the most interesting theoretical minds in neurobiology today. This book summarizes his ideas, but in a somewhat confusing way. In fact, it is really two books. The first and last

chapters are philosophical in nature, aiming to deal with free will, causality, and meaning. These problems are inherently so baffling or ill-posed that no one has discovered a way to build consensus. Arguments about free will cycle over and over again through the centuries, and Freeman can cite Aristotle and Aquinas as easily as more recent philosophers such as Merleau-Ponty. I found this part unsatisfying. It is either too long or too short, and the philosophical discussion by itself is simply unlikely to have lasting impact. Freeman's philosophy is best seen as the heuristic assumptions of a distinguished scientist.

The second book within this volume is a tutorial introduction to Freeman's theoretical neurobiology. All of the ideas are rigorously stated, going step by step from single neurons to dynamical networks. The development is quasi-mathematical, with assumptions and theorems, all stated with great simplicity and clarity. The points are illustrated with very simple tutorial diagrams. This is solid theory and very thought provoking; it deserves to be widely read. I highly recommend it to any scientist interested in brain theory. Unfortunately, it is sandwiched between the philosophical chapters, so that casual readers might well miss it.

Freeman's scientific thought is based on some 50 years of laboratory work on the electrophysiology of rabbit and other mammal brains, starting with studies of olfaction. It begins with two surprising observations: first, a given odor stimulus does not create a fixed electrical field pattern. Rather, activity patterns of the brain are constantly changing and reforming in relation to each other. Second, a repeated stimulus does not elicit the same pattern every time it is presented. Freeman takes these observations to falsify a simple computationalist view of the brain.

What is the alternative? Here, Freeman begins with a fundamental consideration of projection (pyramidal) neurons of the cortex, and their two classical potentials, axonal pulses and dendritic waves. This is familiar territory, but Freeman goes beyond the neuron doctrine to develop a view of the nonlinear properties of neuronal populations with high interactivity, what he calls the "mesoscopic" level of analysis. Neuronal populations are open thermodynamic systems with many semiautonomous elements, each with weak interaction with many others, nonlinear input-output relations, as well as massive energetic and molecular inputs and a limitless sink for waste products and heat. Above a certain connectivity threshold, Freeman argues, neuronal populations change via state transitions. Populations shape the activity of single neurons, but they are not a simple additive result of local activity.

Ten "building blocks" follow from these assumptions (p 35). For example, the first defines a state transition of an excitatory population from a zero to a nonzero point attractor with steady-state activity. The second shows the emergence of oscillations through negative feedback between excitatory and inhibitory populations. The third concerns the state transition from a point attractor to a limit cycle attractor that regulates steady-state oscillation. And so on, all the way to building block ten, the formation of a sequence of global amplitude modulation patterns of chaotic activity that directs the intentional (goal-directed) state of an entire hemisphere. It is an admirable exposition with many attractive features; whether it is true, ultimately, is an empirical question for the future.

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THE AGING BRAIN, *Maps of the Mind.*

By Lawrence Whalley. New York: Columbia University Press. \$22.95. ix + 182 p; ill.; index. ISBN: 0-231-12024-9. 2001.

If we had a choice of avoiding any one of the inevitable consequences of aging, I think most of us would chose declining memory and cognitive ability. They are the consequences of aging that most cause us to be judged and to feel threatened by our youthful predecessors. Following the "decade of the brain," we now have a greatly enlightened view of normal brain function and changes that occur in the brain with age. For example, a prior dogma of brain aging and cognitive decline was that it is due to cell loss and shrinkage in regions of the brain, such as the hippocampus, that are responsible for learning and memory function. We now know that these effects cannot entirely explain memory and cognitive decline with age, inviting the exciting possibility that the decline of certain mental functions with age can be slowed and perhaps even reversed.

*The Aging Brain* provides an excellent, comprehensive overview of our current understanding of the changes that occur in the brain both with age and in age-related diseases. This book not only covers the pathological and biochemical changes that occur with age, and in conditions such as stroke and Alzheimer's disease, but it also addresses the social and psychological impacts of aging on mental function and vice versa. For example, Whalley addresses the important issue of how stress and stressful life events can impact on brain function as we age and how such exposures can increase the risk for age-related disorders of the brain. This book emphasizes the complexity of brain aging, determined by genetic, environmental, and other

factors resulting in much variability in the success of brain aging between individuals. In the final chapter, Whalley covers important considerations for lifestyle and therapeutic interventions for slowing or reversing the effects of brain aging. This book will be of great value to medical and graduate students, as well as anyone who desires an overview of brain aging.

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SELF-ORGANIZING MAP FORMATION: FOUNDATIONS OF NEURAL COMPUTATION, *Computational Neuroscience.*

Edited by Klaus Obermayer and Terrence J Sejnowski. A Bradford Book. Cambridge (Massachusetts): MIT Press. \$34.95 (paper). xvii + 440 p; ill.; index. ISBN: 0-262-65060-6. 2001.

A USER'S GUIDE TO THE BRAIN: PERCEPTION, ATTENTION, AND THE FOUR THEATERS OF THE BRAIN.

By John J Ratey. New York: Pantheon Books. \$27.50. vii + 404 p; ill.; index. ISBN: 0-679-45309-1. 2001.

With its catchy title, this book has several possible audiences—brain scientists, health care professionals, and their patients. Ratey builds a model of mind and mental disorders by using research results, stories of patients, and speculations. He takes a strongly biological view of the underlying causes for such diverse conditions as autism, attention deficit disorders, obsessive-compulsive disorder, depression, social ineptness, and dyslexia, and suggests treatments that reflect the biological brain. I am sympathetic with Ratey's biological approach to mental disorders, but am not convinced that he always has the details right. His claims that specific disorders were produced by deficits in particular brain pathways or neurotransmitters, although reasonable, were not always backed by sufficient evidence. These speculations, however, made for nice stories and allowed Ratey to describe the various pathways and locations in the brain that are thought to be associated with particular mental functions. The author obviously has a grasp of a variety of brain functions and disorders. He did a good job of pointing out the multiple uses of some brain areas, such as the role of the cerebellum not only in coordination of movement, but also in cognition. This book is said to be a condensed and more popularized version of a longer, unpublished work. In creating the volume, Ratey received help from a number of colleagues and from a ghostwriter, Mark Fischette.